$$s(a,b) = \begin{cases} 1 & \text{if } a=b \\ \frac{c}{I(a)I(b)} \sum_{i \in I(a)} \sum_{j \in I(b)} s(i,j) & \text{o.w.} \end{cases}$$

2nd Mar 2026 | CS6370

- Sim Rank → algo. used to measure structural similarity btw. the nodes in a graph. Two obj. are considered similar if they are ref. by similar objects. Eg. Link analysis, etc.
- WordNet → large lexical database of English. Words are grouped into set of synonyms c/d synsets. It tries to establish semantic relationships.
- Introspective methods can only solve the circularity problem partially only due to fundamental limits of the language.
- WordNet → is a lexical resource. It helps to establish connecⁿ btw form & meanings. It has synonyms. Synsets are related to each other with diff. kinds of relationships. It does not solve anything. It just stands as a resource. Corresponding to every word, we get corresponding synset IDs.
- WSD (Word Sense Disambiguaⁿ) - It is a task of determining which particular meaning of a word is being used in the current context.

- Brown Corpus – one of the earliest & most influential corpora in computational linguistics. Acts as founda^n for many NLP tasks.
- SemCor – sense-tagged English corpus where the words have been annotated with their meaning (senses) coming from WordNet. Gold std. dataset for WSD problem.
- IAA (Inter-Annotator Agreement) – It measures how much diff. annotators agree while labelling the same data. Higher the IAA means that the task is reliable.

- Part-of-Speech Tagging – Task of assigning each word in a sentence to its correct grammatical category like noun, verb, etc.

- Watch the WordNet lecture
- Page Rank

$I_1 =$ all page it pt. to. (incoming impt.)

$= \alpha_1 I_1 + \alpha_2 I_2 + \dots$

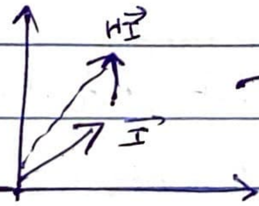
$I_2 =$ ______

$I_m =$ ______

take all of them to form

$I = HI$

H comes from the struc. of web.
We are solving the eigen equa^n.



→ characterizes the transforma^n of the vector.

$\boxed{HI = \lambda I}$ → solved using power method.

when $\lambda = 1$ then we get it.